

Measurement of Self Inductance by Maxwell Bridge

AIM:

To determine the self-inductance of an unknown coil.

Theory

This bridge circuit measures an inductance by comparison with variable standard Self Inductance. The connections for balance condition are shown in Fig. 1.

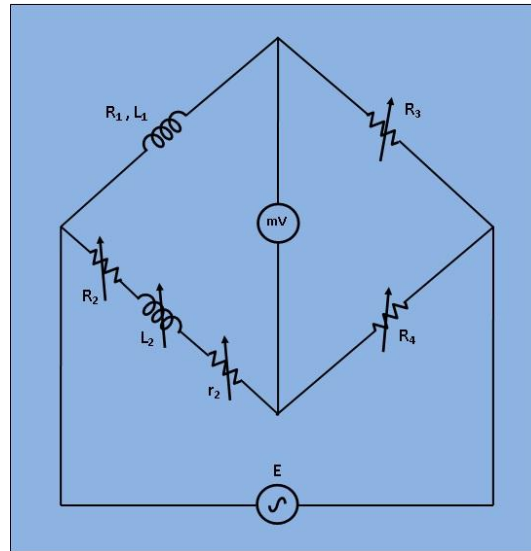


Fig 1: Circuit Diagram for Measurement of Self Inductance by Maxwell Bridge

Let, L_1 = Unknown Self Inductance of resistance R_1 ,

L_2 = variable inductance of fixed resistance r_2 ,

R_2 = variable resistance connected in series with inductor L_2 ,

R_3, R_4 = known non inductive resistances,

At balance condition,

$$(R_1 + j\omega L_1) * R_4 = (R_2 + r_2 + j\omega L_2) * R_3 \dots (1)$$

Equating both the real and imaginary parts in eq (1) and separating them,

$$L_1 = \left(\frac{R_3}{R_4}\right)L_2 \dots (2)$$

$$R_1 = \left(\frac{R_3}{R_4}\right) * (R_2 + r_2) \dots (3)$$

Resistors R_3 and R_4 are normally a selection of values from 10, 100, 1000 and 10,000 Ω . r_2 is a decade resistance box.

Procedure:

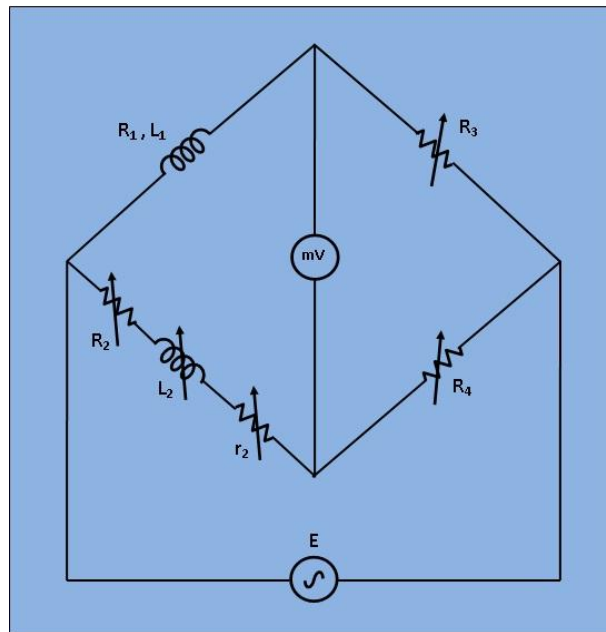


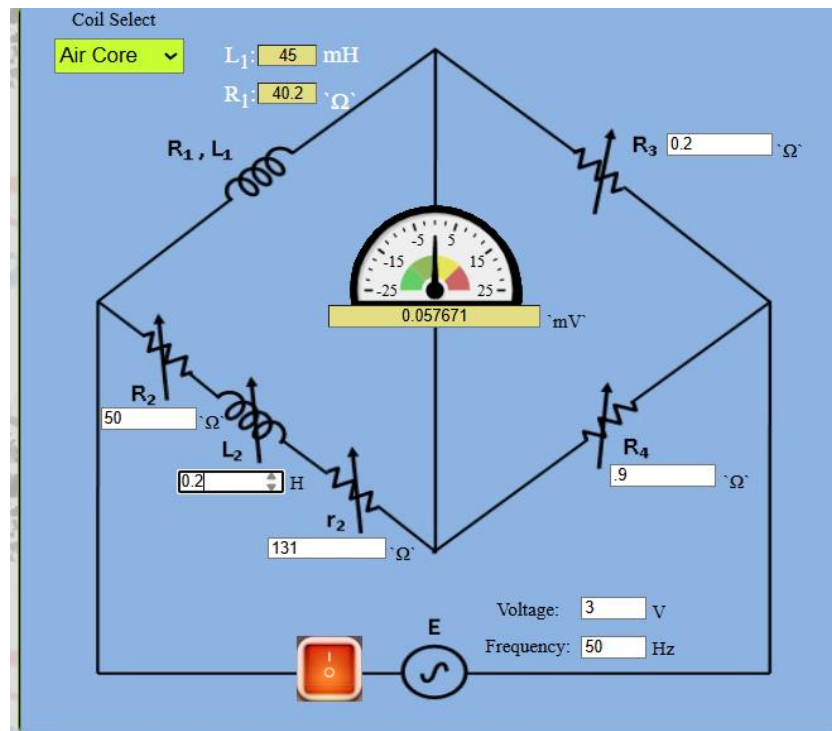
Fig 1: Circuit Diagram for Measurement of Self Inductance by Maxwell Bridge

1. Apply Supply voltage from the signal generator with arbitrary frequency. ($V = 3\text{V}$). Also set the unknown Inductance value from 'Set Inductor Value' tab.
2. Then switch on the supply to get millivoltmeter deflection.
3. Choose the values of L_2 , r_2 , R_2 , R_3 and R_4 from the inductance and resistance box. Vary the values to some particular values to achieve "NULL".
4. Observe the millivoltmeter pointer to achieve "NULL".
5. If "NULL" is achieved, switch to 'Measure Inductor Value' tab and click on 'Simulate'. Observe the calculated values of unknown inductance (L_1) and its internal resistance (R_1) of the inductor.
6. Also observe the Dissipation factor of the unknown inductor which is defined as

$$\frac{\omega L}{R} \text{ Where, } \omega = 2\pi f$$

Simulation:

CASE 1(Air Core):



The current voltmeter reading is: 0.057671 mv.

Now click on simulate to get:

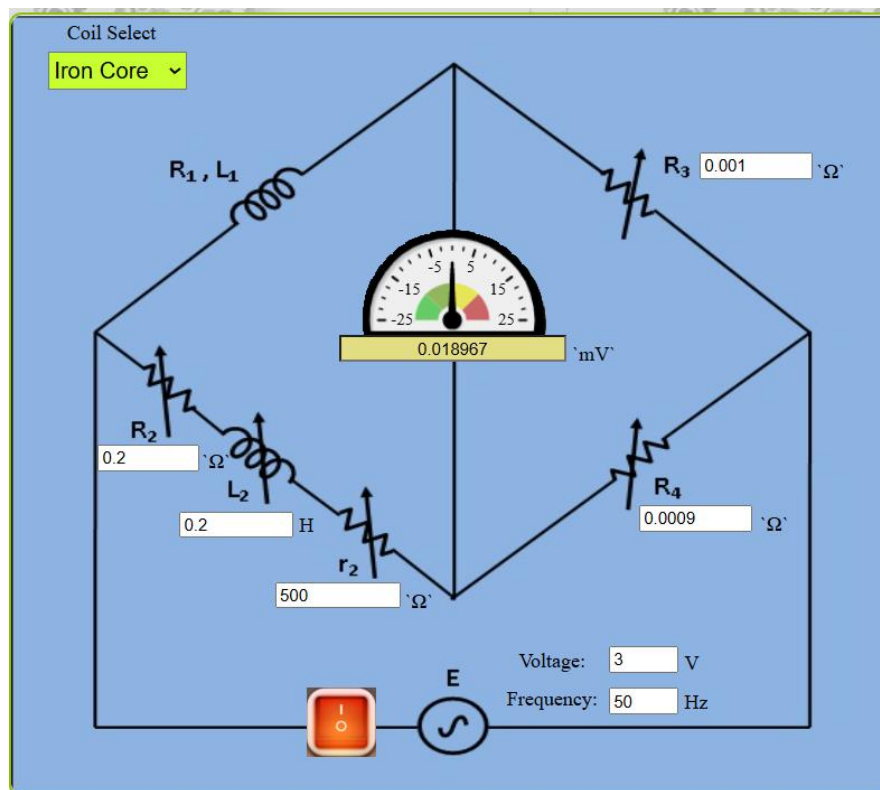
Inductor value (in mH): 44.444444444444

Resistance value (in Ohm): 40.222222222222

Quality Factor: 0.34696

Simulate

CASE 2(Iron Core):



The current voltmeter reading is: 0.018967 mv.

Now click on simulate to get:

Inductor value (in mH): 222.222222222222

Resistance value (in Ohm): 555.777777777777

Quality Factor: 0.12555

Simulate

RESULT:

Thus, the unknown inductance is found using Maxwell Bridge.